

# Early Detection of Autism Spectrum Disorder in Toddlers Using DL Approaches

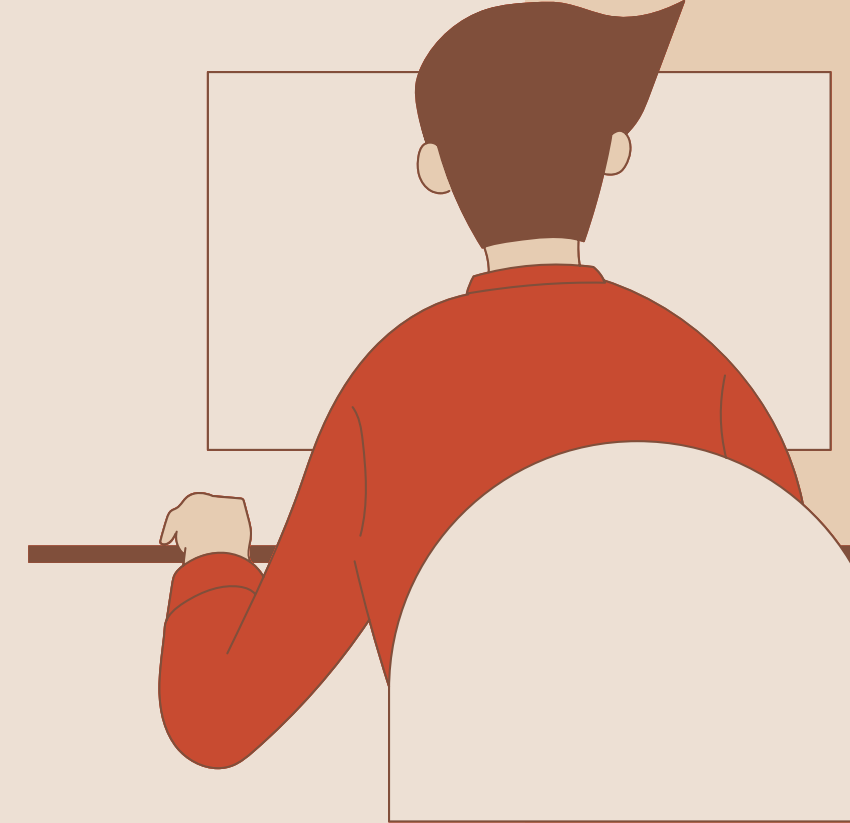
Authors :

Anchal Gupta, Aastha Basu, Aman Kumar, Amar Singh  
Vidyashilp University, Bangalore, India

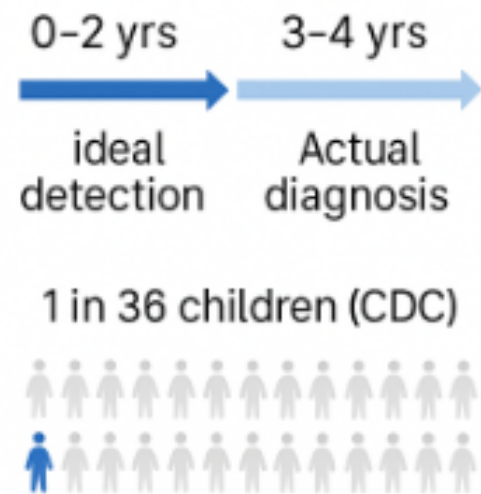
# Table of Contents

|      |                      |    |
|------|----------------------|----|
| I    | Introduction         | 3  |
| II   | Problem Statement    | 4  |
| III  | Literature Review    | 5  |
| IV   | Proposed Methodology | 7  |
| V    | Explainability       | 11 |
| VI   | Experimental Setup   | 12 |
| VII  | Results              | 15 |
| VII  | Discussion           | 17 |
| VIII | Conclusion           | 20 |

# Introduction



## Early Detection Matters



## What is Autism Spectrum Disorder (ASD)?

- A neurodevelopmental disorder affecting social interaction, communication, and behavior
- Highly heterogeneous — varies from child to child

## Why Early Detection Matters

- Early signs appear before age 2
- But average diagnosis happens at 3–4 years
- Missed intervention window = poorer developmental outcomes

## Challenges in Current Diagnosis

- Traditional methods are subjective, slow, and expert-dependent
- Rural or low-resource settings lack access

## The Need

- Automated, objective, and scalable screening systems are urgently needed

## Challenges in Current Diagnosis

Subjective, slow, and expert-dependent



## Need for Automated Screening

Objective and scalable tools

# Problem Statement



## Delayed Diagnosis

- ASD signs appear by 12–18 months, but diagnosis often occurs at 3–4 years
- Critical early intervention window is missed



## Subjective and Expert-Dependent

- Relies on parent recall and expert behavioral observation
- Outcomes vary by clinician training and availability
- Tools like M-CHAT are not standardized globally

## Limited Access in Low-Resource Settings

- Shortage of developmental pediatricians
- Screening tools often proprietary or costly
- Rural areas lack infrastructure for consistent diagnosis

## Need for a Better Approach

- Goal: Two-phased, AI-powered pipeline
  - Phase 1: Parent-led questionnaire
  - Phase 2: Video-based behavioral analysis



# Literature Review – Existing Work

## ML/DL in ASD Detection

- Growing interest in ML/DL for early ASD screening
- Focus on automating behavior analysis and minimizing subjectivity

## Data Modalities Used

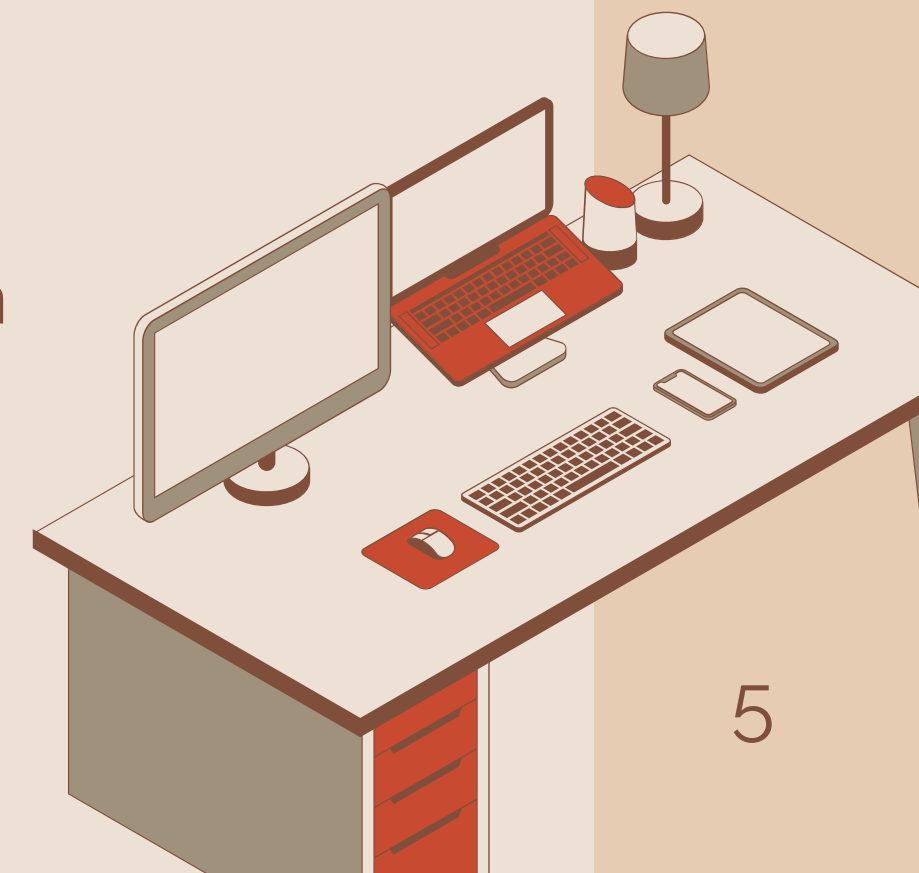
- Questionnaires (e.g., M-CHAT, ASQ)
- Images & Video (facial cues, motor behavior)
- Audio/Speech (response to name, vocal tone)
- Neuroimaging (fMRI, EEG)

## Popular Architectures

- CNNs (MobileNet, ResNet) – for spatial features from video/images
- LSTM/Bi-LSTM – for temporal behavior (e.g., gaze shifts, RTN)
- Hybrid Models – CNN-LSTM for spatiotemporal analysis
- GNNs, Transformers – advanced structures for meta-connectivity, motion

## Limitations in Existing Work

- Lack of Explainability (Black-box models → low clinician trust)
- Dataset Biases (small, homogeneous datasets)
- Poor Real-World Validation (trained/tested only on lab datasets)



# Research Gaps in Early ASD Detection

## Delay in Intervention

- Diagnosis usually happens at 3–4 years, missing critical early development windows (before age 2).
- Results in delayed access to therapy and support.

## Limited Dataset Diversity

- Existing datasets are small, cross-sectional, and lack demographic variety.
- Models trained on them don't generalize well across cultures or regions.

## Poor Multimodal Fusion

- Current systems often treat video, audio, and questionnaire data separately.
- Ineffective fusion methods miss complex behavioral patterns.

## Lack of Real-World Validation

- Most models tested only in lab settings with clean, curated data.
- Uncertain performance in real clinical or home environments.

## Privacy and Ethical Concerns

- Use of sensitive child behavior data raises ethical red flags.
- Lack of privacy-preserving methods (e.g., federated learning, informed consent).



# Proposed Methodology

# System Overview :

## Two-Phase Diagnostic Pipeline

The proposed system introduces a two-stage diagnostic pipeline for early detection of Autism Spectrum Disorder (ASD) in toddlers. It mirrors real-world clinical workflows, combining subjective caregiver input with objective behavioral analysis, ensuring both scalability and clinical fidelity.

### Phase 1: Questionnaire-Based Screening

Clinically grounded in DSM-5, AAP, CDC, and NIMH guideline

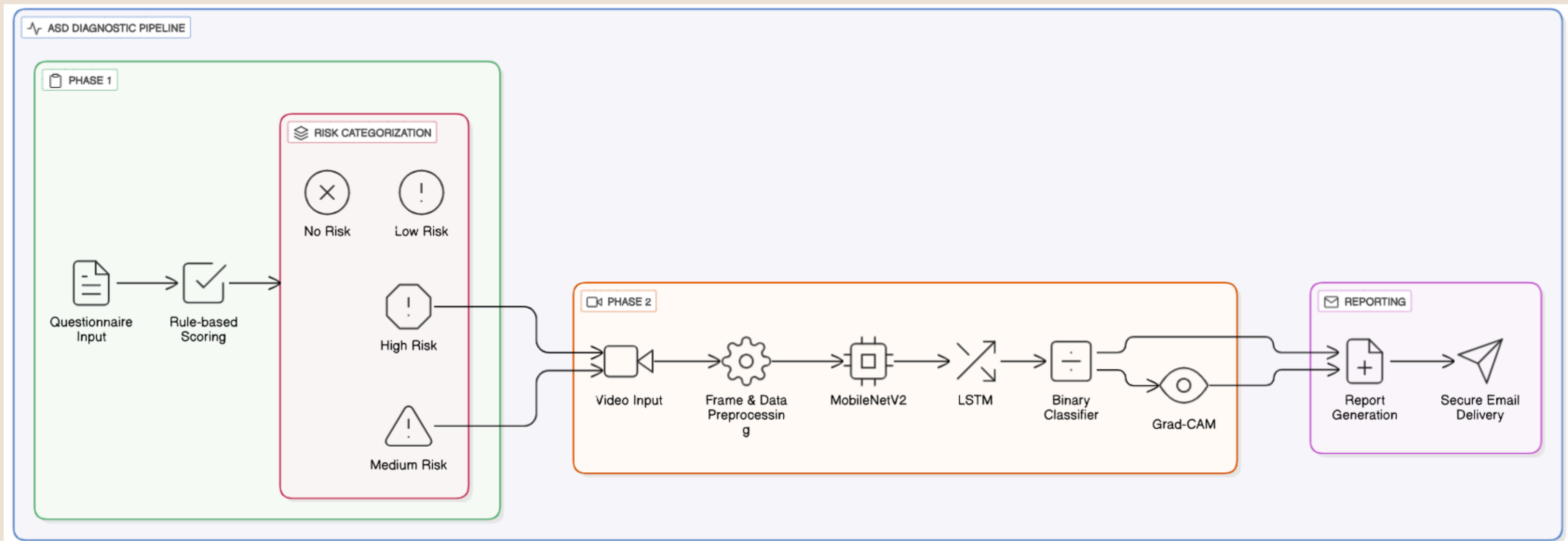
Parent-administered behavioral questionnaire.

- Social interaction & communication
- Play & imagination
- Behavioral/sensory sensitivities
- Developmental regression
- Family history

Responses evaluated via a rule-based red flag system.

Scoring system:

- Each Yes/No mapped to “red flag” criteria.
- Risk categories based on red flag count:
  - No Risk: < 4
  - Low Risk: 4–7
  - Medium Risk: 8–14
  - High Risk:  $\geq 15$
- Toddlers with Medium or High Risk are referred to Phase 2 (video analysis).



**Figure 1 : Overall architectural diagram of the two-phased ASD diagnostic pipeline, illustrating the questionnaire-based screening (Phase 1), video-based behavioral analysis (Phase 2), Grad-CAM explainability, and automated report generation.**

# System Overview :

## Two-Phase Diagnostic Pipeline

### Phase 2: Video-Based Behavioral Analysis

#### Input Videos

- 30–60 seconds of toddler behavior in natural/semi-controlled settings
- Focus: capturing gaze, gesture, facial expression, interaction

#### Preprocessing

- Frame extraction at 10 FPS
- Resize to 224×224 pixels
- Normalization using ImageNet stats
- Augmentation: flip, contrast, light rotation (only on training set)

#### Model Architecture: CNN-LSTM

- CNN (MobileNetV2): Extracts spatial features per frame
- LSTM: Captures sequential behavioral patterns over time
- Final layers: Dropout + Fully Connected for binary classification

#### Output

- Prediction: ASD / Non-ASD
- Enables objective behavioral screening

# Final Output : Actionable Results & Reports

## What the System Produces

- Risk classification (No / Low / Medium / High)
- ASD prediction (Positive / Negative)
- Video analysis result (e.g., "Repetitive hand movement detected")
- Grad-CAM visualizations showing focus areas

## Data Privacy

- Secured transmission (email encryption + consent)
- Designed for quick, trusted communication in low-resource settings

## Caregiver Report

Delivered via email with:

- Risk category
- Red flag summary
- Final Prediction
- Video Analysis
- Reasoning

# Explainability – Grad-CAM

## What is Grad-CAM?

- Gradient-weighted Class Activation Mapping (Grad-CAM) highlights which areas in video frames influenced the model's prediction.
- Applies to final CNN layers (e.g., MobileNetV2) to generate heatmaps on important regions.

## What it Shows:

- Warm colors (yellow/red) indicate key attention regions (e.g., hands, face, torso).
- Helps explain behavior-based decisions such as:
  - Repetitive hand movements
  - Gaze aversion
  - Unusual facial expressions

## Why It Matters (Clinical Impact):

- Reduces the "black box" problem of deep learning.
- Enables clinicians and caregivers to understand why a prediction was made.
- Aligns predictions with known behavioral markers (hand flapping, lack of eye contact).
- Improves:
  - Diagnostic trust
  - Caregiver communication
  - Ethical transparency

# Experimental Setup

# Data Collection



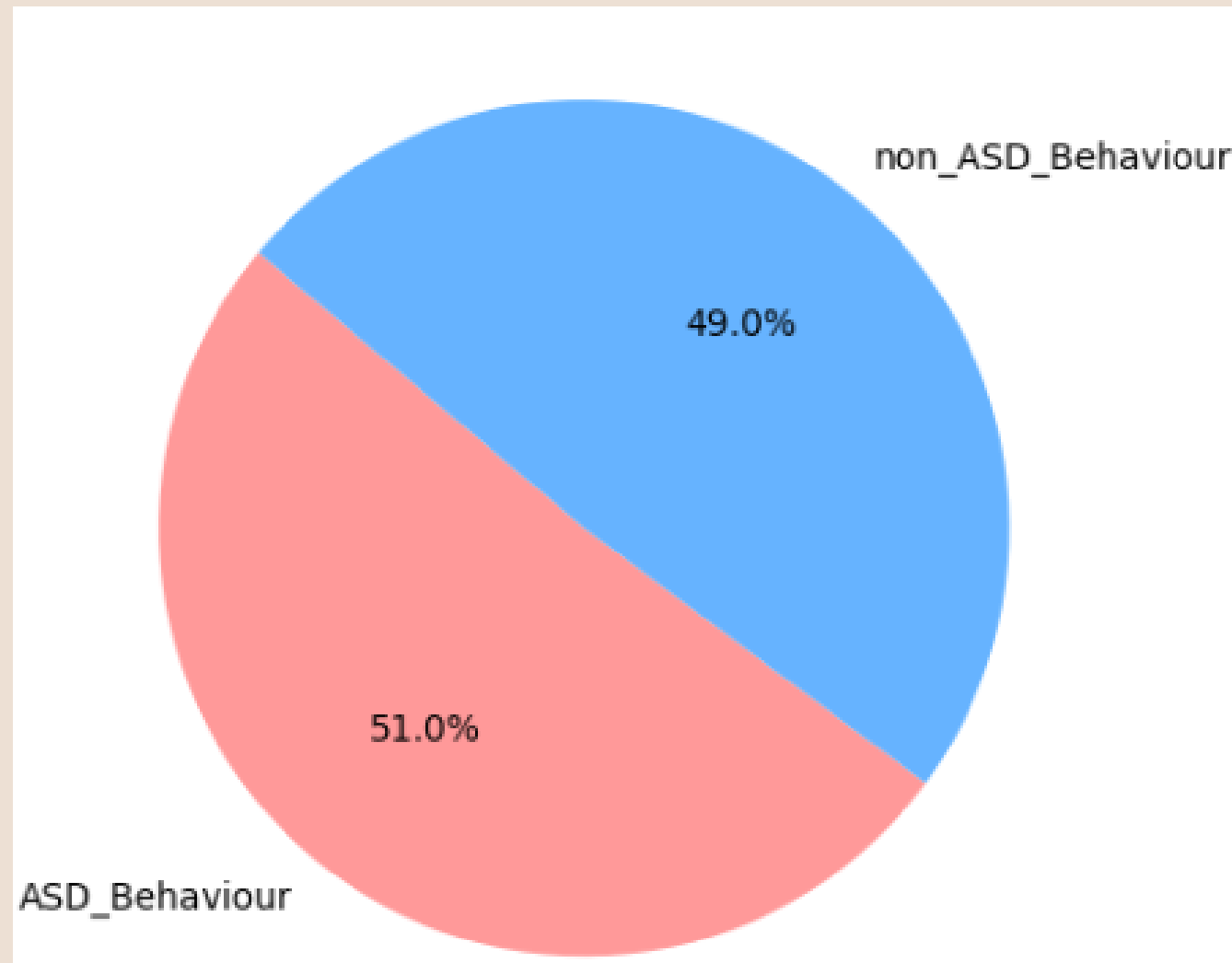
## Questionnaire Data

- Custom-designed screening tool based on clinical standards:
- DSM-5, CDC, AAP, and NIMH.
- Parents respond to structured Yes/No questions across 5 key domains:
  - Social interaction & communication
  - Play & imagination
  - Behavioral patterns & sensory sensitivities
  - Developmental history & regression
  - Family history of ASD
- Responses mapped to “red flag” indicators and classified into:
  - No Risk (<4 flags)
  - Low Risk (4–7 flags)
  - Medium Risk (8–14 flags)
  - High Risk (15+ flags)

## Video Data

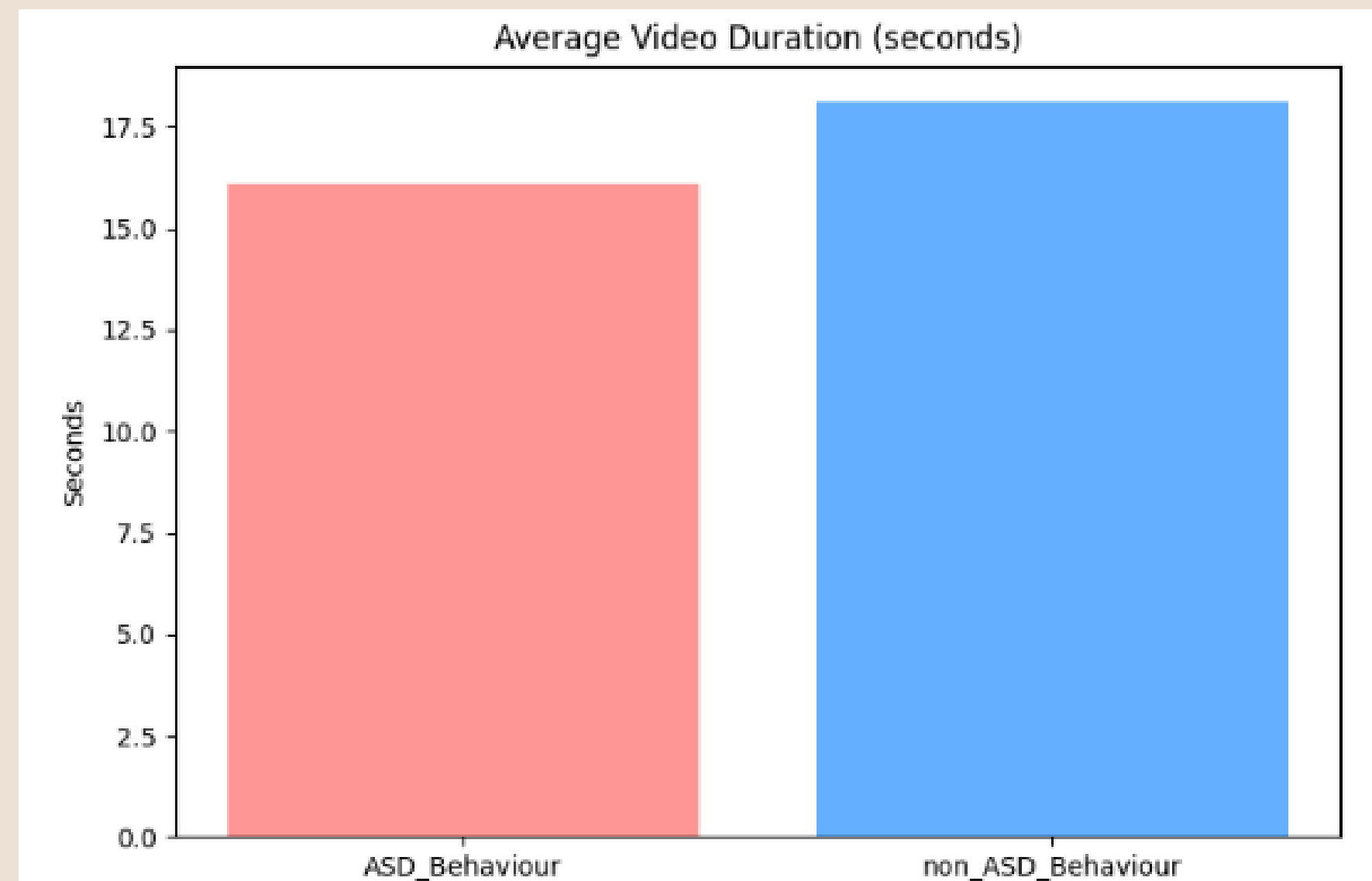
- ASD Class:
  - Sourced from the Social Signal Behavior Database (SSBD)
  - Over 10 hours of annotated videos of toddlers (ages 2–4) diagnosed with ASD.
- Non-ASD Class:
  - Publicly available YouTube Shorts with typically developing toddlers.
  - Inclusion criteria:
    - Clear visibility of toddler in a natural setting
    - No copyright restrictions or facial blurring
    - Age range: 2–4 years
    - Public domain content only
- Ensured balanced class distribution for training and evaluation.

# Data Collection



**Figure 2:** Distribution of Videos in each Category

**Figure 3:** Average Duration of Videos in each category in seconds



# Model Training



## Model Architecture (Phase 2 – Video Classification)

- CNN-LSTM hybrid model:
  - CNN Backbone: Pretrained MobileNetV2 for spatial feature extraction from video frames.
  - LSTM Head: Single-layer LSTM with 64 hidden units to capture temporal behavior (e.g., gaze, repetitive motion).
  - Final hidden state passed through:
    - Dropout ( $p = 0.5$ )
    - Fully connected layer for binary classification: ASD / Non-ASD

## Training Setup

- Dataset split:
  - 70% Training, 15% Validation, 15% Test (stratified for class balance)
- Loss function:
  - Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss)
- Optimizer:
  - Adam optimizer
  - Learning rate: 0.0001
  - Weight decay:  $1e-5$
- Epochs:
  - Max 18 with early stopping
  - Patience = 5 epochs without validation loss improvement

# Results



| <b>Metric</b>                   | <b>Model 1 (86%)</b> | <b>Model 2 (90%)</b> | <b>Model 3 (94%)</b> |
|---------------------------------|----------------------|----------------------|----------------------|
| <b>Accuracy</b>                 | 85.71%               | 90.48%               | 93.75%               |
| <b>Precision</b>                | 1.00                 | 1.00                 | 1.00                 |
| <b>Recall</b>                   | 0.77                 | 0.82                 | 0.88                 |
| <b>F1-score</b>                 | 0.87                 | 0.90                 | 0.93                 |
| <b>ROC-AUC</b>                  | 0.99                 | 0.98                 | 0.92                 |
| <b>Best Validation Accuracy</b> | 94.12%               | 88.24%               | 100.00%              |

# Sample Output & Report



## Assessment Report for {{ child\_name }}

Dear {{ parent\_name }},

Thank you for completing the autism screening assessment. Here is a summary:

**Risk Category:** {{ questionnaire\_risk\_category }}

**Red Flags Count:** {{ final\_red\_flags\_count }}

**Final Prediction:** {{ final\_prediction }}

**Video Analysis:** {{ video\_analysis\_outcome }}

**Reasoning:** {{ video\_analysis\_reason }}

If you have any questions or would like to speak to a professional, please consider consulting a pediatrician or developmental specialist.

Best regards,

*Autism Screening Team*

- Personalized caregiver report generated after screening
- Includes key insights from questionnaire and video phases
- Risk category (No / Low / Medium / High)
- Count of behavioral red flags detected
- Final AI prediction: ASD / Non-ASD
- Short behavioral reasoning summary (e.g., "Detected gaze aversion")
- Delivered via email for quick follow-up and early action

# Discussion



# System Benefits



## Clinically Aligned

- Mirrors real-world clinical workflows: Phase 1 = parent-based screening, Phase 2 = observation-based evaluation.
- Follows DSM-5, AAP, and CDC screening guidelines.
- Increases trust among pediatricians and caregivers.

## Resource-Efficient

- Lightweight models like MobileNetV2 and rule-based Phase 1 ensure low computational cost.
- Only Medium/High risk toddlers undergo video analysis, saving time and processing power.

## Explainable, Interpretable AI

- Uses Grad-CAM to show visual heatmaps (e.g., focus on hand-flapping or gaze).
- Clinicians and caregivers can understand why the AI flagged a child.
- Reduces the “black box” concern in medical AI.

## Real-Time Automated Screening

- Rapid end-to-end workflow from questionnaire to video assessment.
- Caregivers get automated reports via email with detailed results.
- Enables faster referral to developmental specialists if needed.



# Limitations and Challenges



## Limited Real-World Testing

- System validated only in controlled settings with curated datasets.
- Lacks evaluation in clinical or community-based environments.

## Email Report Delivery Challenges

- Dependence on digital literacy, email reliability, and internet connectivity.
- Risk of miscommunication, delivery failure, or data breach.

## Ethical Concerns (Consent & Privacy)

- Use of public toddler videos may raise ethical issues regarding informed consent.
- Sensitive health data handling requires adherence to privacy standards (e.g., GDPR, HIPAA).

## Small, Homogeneous Dataset

- Dataset lacks diversity in terms of ethnicity, geography, and socioeconomic background.
- Reduces the generalizability of the model across populations.

## YouTube Video Quality Variance

- Non-ASD data from public videos may differ in lighting, background noise, camera angles.
- Inconsistent quality may introduce bias or reduce model reliability.



# Future Work



**Expand dataset with diverse samples:** Increase sample size and include data from varied ethnic, linguistic, and socioeconomic backgrounds to improve generalizability.

**Pilot clinical trials in real settings:** Test the system in hospitals, rural clinics, and mobile units to evaluate usability and diagnostic effectiveness outside lab conditions.

**Explore advanced temporal models:** Experiment with architectures like Transformers to better capture complex behavioral patterns over time.

**Engage autistic individuals and families:** Incorporate real feedback from caregivers and the autistic community to align the tool with real-world needs and neurodiversity values.

**Strengthen privacy and security:** Use Federated Learning and Differential Privacy to enable data sharing across centers without compromising user confidentiality.

# Conclusion

- Combined questionnaire and video analysis offers a practical, step-by-step screening approach.
- Achieved strong performance metrics — especially high accuracy and zero false positives.
- System design mirrors real clinical workflows, making it easier to integrate in practice.
- Reduces reliance on specialists and improves access in low-resource or rural areas.
- Automated, explainable outputs help caregivers take action sooner.
- Sets a foundation for scalable, AI-assisted autism screening in early childhood.
- Future validation in real-world settings can further improve reliability and impact.



**THANK YOU**